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Polynomial Regression Report

1. Objective and approach

The objective is to predict a continuous target y using polynomial regression. Var1 predicts the Net Power Score of a steam turbine system from six operating inputs, x_1 to x_6 . Var2 predicts the Thermal Anomaly Score from three spatial inputs, x_1 to x_3 . A separate model is trained for each problem.

Data checks and split

Each problem has 1,000 labeled training rows and 1,000 unlabeled test rows. All supplied values are numeric and finite, with no missing cells. Input values lie between -1 and +1. No rows were deleted or imputed. Var2 has 990 distinct training coordinates; repeated measurements were retained.

The labeled data were split into 800 development rows and 200 holdout rows using a fixed seed of 42, preserving broad endpoint-count proportions. Var2 keeps repeated coordinates together, with no coordinate overlap between development and holdout. The final models are refitted on all 1,000 labeled rows.

Polynomial features and regularization

Polynomial expansion creates powers and interactions of the inputs. Every term has total degree at most d : for example, x_1 squared times x_2 has total degree 3. Regression is linear in the coefficients of these polynomial terms. [1]

$$y_{\text{hat}} = b_0 + \text{sum of } b_j * \text{polynomial_term}_j(x)$$

Polynomial terms are standardized before regression so their scales are comparable. Expansion, scaling and fitting occur within each training fold. Validation labels are not used to learn these transformations. Ridge and Lasso were compared to control large coefficients. Lasso can set unhelpful coefficients to zero, and alpha controls its penalty strength. Both selected models use ElasticNet with L1 ratio 1, which gives the Lasso objective. [2]

Validation and metrics

Five-fold CV trains on four folds and predicts the remaining fold until every development row has one out-of-fold prediction. Var1 uses shuffled KFold. Var2 uses shuffled GroupKFold by the full (x_1 , x_2 , x_3) coordinate, keeping a site's measurements in the same fold. [3] The main repeated comparison averages out-of-fold MSE over ten shuffles, with seeds 100 to 109.

$$\text{MSE} = (1/n) * \text{sum } (y - y_{\text{hat}})^2$$

$$R^2 = 1 - \text{sum } (y - y_{\text{hat}})^2 / \text{sum } (y - \text{mean}(y))^2$$

Lower MSE is better. R squared measures improvement over predicting the evaluation-set mean; values closer to 1 indicate a better fit.

2. Var1: steam turbine power

Search and degree choice

The initial Ridge search checked all 63 nonempty feature subsets. Degrees 1 to 10 were explored, with staged limits on larger subsets to control computation. The all-six-input Ridge search was then extended through degree 10. Sparse polynomial models were tested at degrees 3 to 6.

The graph shows Lasso using all six inputs and a fixed alpha of 0.01. The same shuffled five-fold split was used for every point. MSE falls from degree 3 to degree 5, then rises at degree 6. Degree 5 has the lowest error in this comparison.

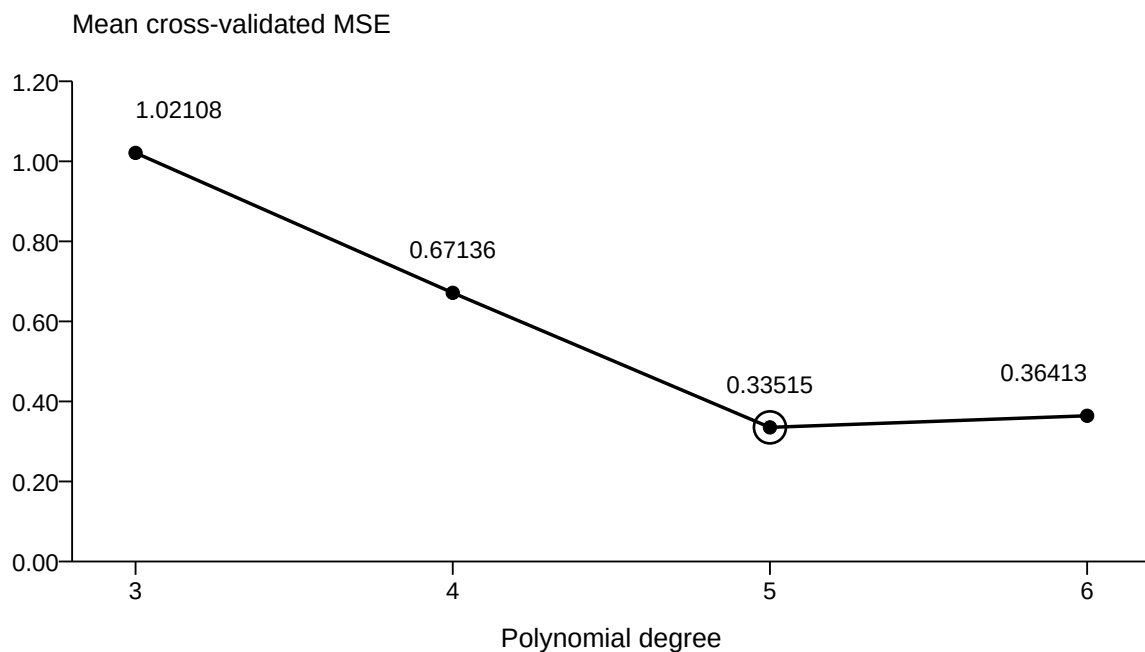


Figure 1. Var1 MSE versus degree: all six inputs, Lasso alpha 0.01, five-fold CV with seed 42. The ring marks the selected degree 5.

Final model and results

The final model is **degree-5 polynomial Lasso with alpha 0.01**, using x_1 to x_6 . Its initial five-fold MSE is 0.33515. Ten repeated CV shuffles give mean MSE **0.33705**. Degree-5 Ridge with alpha 10 gives 0.61446 on the same ten repeats. Later local alpha checks did not justify replacing 0.01.

On the existing 200-row holdout, MSE is **0.31909** and R squared is **0.96959**. These scores describe an 800-row fit and are diagnostic because this holdout was reused.

The final 1,000-row fit retains 110 nonzero coefficients from 461 available polynomial terms, excluding the intercept. The reduced coefficient set controls complexity while retaining interactions.

3. Var2: thermal reservoir mapping

Search and degree choice

All seven nonempty feature subsets were checked. Ridge degrees 1 to 20 and computationally manageable ordinary least-squares models were screened, followed by Lasso and Elastic Net refinement. Higher degrees and a Legendre polynomial basis were also checked. All three original inputs were retained.

Degree 12 was the initial Lasso choice. The later repeated grouped-CV comparison favored degree 10. The graph compares the Lasso finalists on the same ten grouped shuffles. Degrees 10 and 12 use alpha 0.001; degree 13 uses alpha 0.0013. Degree 10 has the lowest mean MSE among these compared finalists.

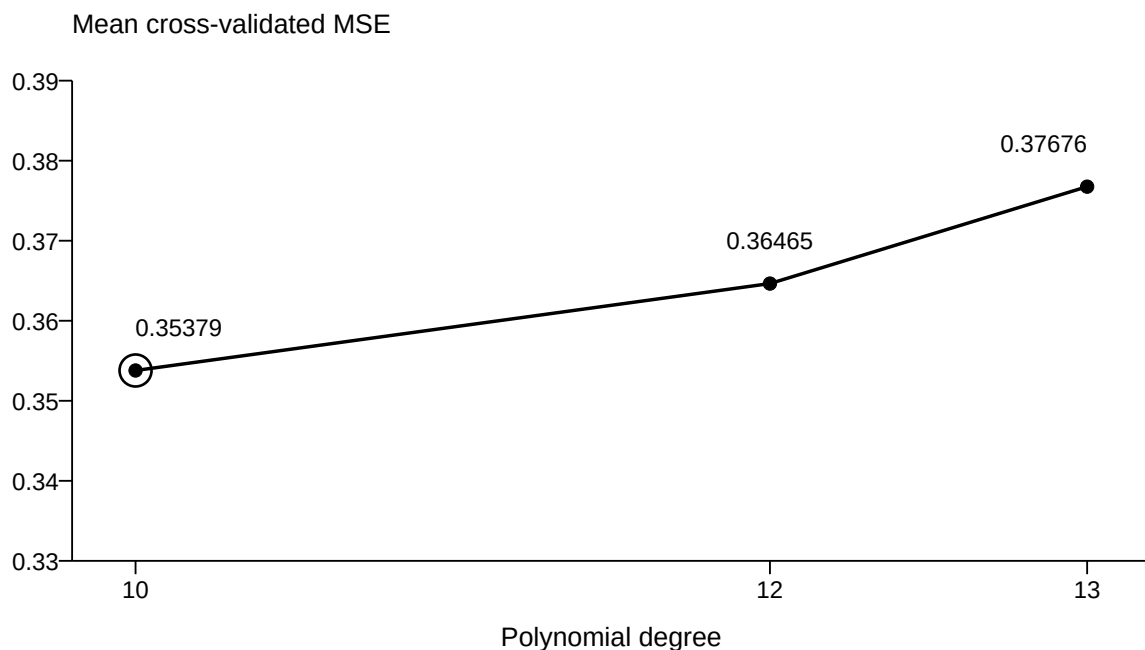


Figure 2. Var2 mean repeated-CV MSE versus degree for the compared Lasso finalists. The ring marks degree 10. The vertical scale focuses on 0.33 to 0.39; the improvement is modest. This is not an exhaustive repeated-CV degree sweep.

Final model and results

The final model is **degree-10 polynomial Lasso with alpha 0.001**, using x_1 , x_2 and x_3 . Its mean repeated-CV MSE is **0.35379**, compared with 0.36465 for degree 12. Degree 10 has approximately 3% lower average MSE and wins six of ten matched repeats. This supports the final choice without establishing a guaranteed minimum hidden-test error.

On the existing 200-row holdout, MSE is **0.23984** and R squared is **0.99496**. These are diagnostic scores from an 800-row fit, because the holdout was previously evaluated for degree 12.

The final 1,000-row fit retains 134 nonzero coefficients from 285 available polynomial terms, excluding the intercept. Each CV fit uses roughly 640 rows; smaller training sets and different validation composition can explain a higher CV MSE than the holdout MSE.

4. Checks, limitations and final output

Boundary coverage

An endpoint input is exactly -1 or +1. In var1, rows with at least four endpoint inputs make up 9.4% of development data but 37.4% of test data. The test set also has 16 rows with all six inputs at endpoints, while training has none. Boundary weighting was tested at the chosen settings and worsened CV results, so the unweighted model was retained.

In var2, training contains 16 rows with all three inputs at endpoints, and test contains 14. The holdout contains no such rows. Grouped CV withholds corner sites during validation, providing relevant evidence beyond this holdout. It cannot establish the true errors on the test corners.

Test-mix proxy and evaluation limits

A descriptive proxy weights development CV errors by the proportions of endpoint-count groups in the unlabeled test inputs. It uses no hidden test labels. The proxy MSE is **0.34748** for var1 and **0.35411** for var2.

These are rough estimates. Broad groups pool different boundary patterns and do not precisely estimate corner-only error. Both holdouts were reused: three models for var1 and two for var2. Their scores are diagnostic, not untouched test estimates. CV scores were also used for selection, and repeats reuse the same observations. Hidden-test MSE and R squared cannot be calculated without the hidden target values.

Final training and prediction files

The selected models were refitted on all 1,000 labeled rows for each problem. Both final fits met their numerical convergence tolerance. Test inputs receive predictions from these final models. Each output preserves all 1,000 test rows and the original feature strings in their original order, then appends a finite predicted y. Repeated inputs receive consistent predictions. No predictions were clipped or manually corrected.

IMT2024064_pred_var1.csv contains columns x1, x2, x3, x4, x5, x6, y.

IMT2024064_pred_var2.csv contains columns x1, x2, x3, y.

Conclusion

The final choices are degree-5 Lasso with alpha 0.01 for var1 and degree-10 Lasso with alpha 0.001 for var2. They combine polynomial interactions with regularization and validation-based selection. All predictions were generated using the full labeled datasets and checked against the supplied files.

References

- [1] scikit-learn: PolynomialFeatures.
- [2] scikit-learn: Lasso and its coefficient penalty.
- [3] scikit-learn: GroupKFold.